

Personality Change After Traumatic Brain Injury: A Systematic Review

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Citation

Lauren Burns, Keishema Kerr, Vaughan Bell. Personality Change After Traumatic Brain Injury: A Systematic Review. PROSPERO 2024 Available from <https://www.crd.york.ac.uk/PROSPERO/view/CRD42023440990>

REVIEW TITLE AND BASIC DETAILS

Review title

Personality Change After Traumatic Brain Injury: A Systematic Review

Review objectives

What does the current evidence base reveal about personality change after a Traumatic Brain Injury (TBI)?

Every year, approximately 1.4 million patients attend emergency departments following a head injury in England and Wales (NICE, 2016). Traumatic brain injuries (TBI), such as concussions, contusions, brain haemorrhages and intracranial haematomas are one of the major causes of death and disability in people under the age of 40 in the UK (NICE, 2016). The vast majority of individuals appear to present with normal or minimally impaired consciousness level (95%) following such events, the remaining 5% sustain more fatal outcomes. Following a TBI, there is a risk of a variety of physical, psychological, cognitive, and behavioural changes. Such post-injury changes have been examined in the literature, however, to our knowledge, the current literature is lacking a systematic review that compiles research looking at personality changes. Therefore, this review aims to present an overview of the literature exploring personality changes following a TBI. This includes seeking to understand the classification, diagnosis, prevalence, and rehabilitation of personality changes as well as provide an overview of the research designs, methods and outcomes of the literature to date.

SEARCHING AND SCREENING

Searches

Sources

Four databases will be searched electronically:
MEDLINE, PsycINFO, EMBASE and CINAHL.

Citations And bibliographies of articles and reviews of an applicable nature will be hand searched to avoid missing key articles missed in the electronic search.

Search Date

Studies published between July 1980 and July 2023, given the widespread use of neuroimaging for assessment of TBI from the 80s.

Restrictions

- Reviews, second-hand accounts of case studies;
- Studies unavailable in English;
- Animal studies;
- Research that focuses solely on adolescent or paediatric populations;
- Studies that observed other acquired brain injuries.

Study design

The comparators/controls will vary dependant on the study design. This could include within groups designs whereby participant data are compared to pre-morbid TBI data, or between group designs whereby participant data are compared against control groups (treatment as usual, placebo, or matched pairs group). Some studies may be specific case studies therefore no explicit comparator or control.

ELIGIBILITY CRITERIA

Condition or domain being studied

Personality change following traumatic brain injury (TBI).

Population

Inclusion Criteria

To be included into our systematic review, eligible studies must:

- Investigate personality change following a traumatic head injury;
- Be an original research article with unique data;
- Be published in English;
- Be a human study;
- Either solely have participants who are aged 18 years and older, or, if there is a wider age-range, the adult data must be observable independent of those under 18 years old.

Exclusion Criteria

- Reviews, second-hand accounts of case studies;
- Studies unavailable in English;
- Animal studies;
- Research that focuses solely on adolescent or paediatric populations;
- Studies that observed other acquired brain injuries.

Intervention(s) or exposure(s)

Any original studies that explore personality changes following a traumatic brain injury will be included. Any methods for assessing changes can be included.

Comparator(s) or control(s)

The comparators/controls will vary dependant on the study design. This could include within groups designs whereby participant data are compared to pre-morbid TBI data, or between group designs whereby participant data are compared against control groups (treatment as usual, placebo, or matched pairs group). Some studies may be specific case studies therefore no explicit comparator or control.

OUTCOMES TO BE ANALYSED

Main outcomes

1. Personality change following a TBI identified by changes in structured personality scales or

- diagnosis of post-TBI personality disorder;
2. Prevalence of personality change or disorder after TBI;
 3. Quality of research within this area using Joanna Briggs Collaboration structured assessment tools;
 4. Diagnostic or classification tools used to assess post-TBI personality change;
 5. Clinical characteristics associated with personality changes following a TBI as described by study authors or recorded in specific assessment measures;
 6. Treatment modalities described in the literature;
 7. Treatment outcomes for personality changes following a TBI assessed either through changes in validated measures or outcome descriptions in non-systematic studies;
 8. Lesion location associated with post-TBI personality change.

Additional outcomes

None planned.

DATA COLLECTION PROCESS

Data extraction (selection and coding)

Two reviewers will screen titles and abstracts of articles identified through the search strategy and determine eligibility for inclusion. Full texts of included studies will be retrieved and screened by two reviewers. Disagreements across any stage will be discussed with a third reviewer in order to reach a consensus. The appropriate data will be extracted independently from each of the eligible papers using a standardised data extraction form.

Data for extraction:

- 1) Author details (names, year of publication, country of publication);
- 2) Aim;
- 3) Study design;
- 4) Methods (setting, participant characteristics, number of participants, inclusion/exclusion criteria, comparison group, diagnostic and treatment measures);
- 5) Details of TBI (nature, time, pre-morbid history, comparative data);
- 6) Primary outcome;
- 7) Other outcomes.

Risk of bias (quality) assessment

The STROBE adaptation of the Cochrane QUADAS criteria for rating methods quality of observational studies of clinical cohorts will be adhered to throughout this review. The QUADAS-2 tool, which assesses risk of study-level bias, will also be utilised.

PLANNED DATA SYNTHESIS

Strategy for data synthesis

We expect the literature to be quite disparate in terms of the study design, quality of research, and outcomes with few studies reporting systematic outcomes. However, we aim to conduct meta-analysis where the data supports it. We will use a random effects meta-analysis of binary outcomes to create a pooled meta-analytic estimate of prevalence for studies reporting proportion of TBI cases where personality change or personality disorder is present. We will estimate treatment effects from RCTs using a random-effects meta-analysis of continuous outcomes if mean difference or equivalent is reported, and a random-effects meta-analysis of risk ratio difference if categorical outcomes are reported, comparing between intervention and control groups. However, where this approach is not possible, these findings will be synthesised in a narrative manner. Diagnostic and classification tools, clinical characteristics, treatment modalities, and lesion location will be synthesised narratively.

Analysis of subgroups or subsets

None planned

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members

- Lauren Burns, UCL
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- Vaughan Bell, UCL

Review affiliation

University College London

Funding source

No funders sponsored the review.

Named contact

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TIMELINE OF THE REVIEW

Review timeline

Start date: 16 July 2023. End date: 31 October 2023

Date of first submission to PROSPERO

07 July 2023

Date of registration in PROSPERO

18 July 2023

CURRENT REVIEW STAGE

Publication of review results

The intention is to publish the review once completed. The review will be published in English

Stage of the review at this submission

Review stage

Pilot work

Formal searching/study identification

Screening search results against inclusion criteria

Data extraction or receipt of IP

Risk of bias/quality assessment

Data synthesis

Started

Completed

Review status

The review is currently planned or ongoing.

ADDITIONAL INFORMATION

Additional information

There is a similar review recorded on PROSPERO which was completed as part of a Masters' project (https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=177606), it was due to be completed in 2020 however PROSPERO records have not been updated since the initial record. Given the search methods, there is a possibility their review date range is 2015-2020, however ours will run from 1980-2023. There was no evidence of this review being published, our research team seeks to ensure this work is published given its importance.

PROSPERO version history

- Version 1.0 published on 18 Jul 2023

Review conflict of interest

None known

Country

England

Medical Subject Headings

Brain; Brain Concussion; Brain Injuries, Traumatic; Consciousness; Contusions; Craniocerebral Trauma; Hematoma; Humans; Prevalence; Wales

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